

PRITOM MAJUMDER

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RESEARCH INTEREST

- **Functional & Piezoelectric Ceramics:** Lead-free ferroelectrics, solid-state synthesis, dielectric and piezoelectric response, magnetic ceramics and low-temperature sintering routes.
- **Oxide Semiconductor Thin Films:** Doping-driven control of structural, optical, magnetic and electronic behaviour in transparent conducting oxides.
- **Photonic & Optoelectronic Materials:** Advance ceramics, lasers, LEDs, photovoltaics, photonics and magnetic data storage and optical fibers.
- **Metallurgy:** Corrosion and surface engineering, ferrous metallurgy, materials characterization and mechanical behaviour of materials.

EDUCATION

B.Sc. in Materials and Metallurgical Engineering

2021 to June 2026

Bangladesh University of Engineering and Technology (BUET), Dhaka

CGPA: 3.64 (15th of 59) / 4.00

Relevant coursework: Advanced Ceramics (A+); Nanostructured Materials & Thin Films (A+); Electrical, Optical and Magnetic Properties of Materials (A+); Characterisation of Materials Laboratory (A+); Photonic and Magnetic Materials (A); Characterisation of Materials (A-); Crystallography and Structure of Materials (A-); Phase Diagrams & Transformations (A-).

ON GOING WORK

1. **Pritom Majumder**, Md. Atik Faisal, Mehrab Rahman, Anabil Dey, and Md. Azizar Rahman. "Structural, Optical, Magnetic, and Electronic Properties of Hydrothermally Grown Cu-Doped SnO₂ Thin Films: A Combined Experimental and First-Principles (DFT+U) Study." Manuscript in preparation.
2. **Pritom Majumder**, Anabil Dey, and Ahmed Sharif. "Advancements in Cold Sintering of Piezoceramics: Flux Mediation, Microstructural Evolution, and Enhanced Dielectric Properties - A Review." Manuscript in preparation.

RESEARCH EXPERIENCE

Undergraduate Thesis Researcher · [Ahmed Sharif's Lab](#) · Department of MME, BUET

2025 to 2026

"Investigation of Dielectric and Piezoelectric Properties of KNN with BCHfT Composite Ceramics" ; Supervisor: Prof. Ahmed Sharif ; [THESIS BOOK & THESIS PPTX](#)

- Synthesized lead-free K_{0.5}Na_{0.5}NbO₃ (KNN) ceramics modified with Ba_{0.85}Ca_{0.15}Hf_{0.1}Ti_{0.9}O₃ (BCHfT) by the conventional solid-state route.
- Characterized phase evolution and microstructure by XRD and FE-SEM; measured frequency- and temperature-dependent dielectric permittivity and loss, impedance spectroscopy, and the piezoelectric charge coefficient d₃₃.
- Identified KNN-1wt% BCHfT as the optimum composition, giving the best balance of dielectric response.

Undergraduate Thesis Researcher · [Solid State Physics Laboratory](#) · Department of Physics, BUET

2024 to 2026

- Synthesis of SnO₂ and Cu:SnO₂ research, Characterisation: PL, UV-Vis, I-V, C-V, Mott-Schottky.

Undergraduate Thesis Researcher · [Pilot plant and Process Development Center](#) · BCSIR

2025 to 2026

- Carried out the complete characterization of the KNN-BCHfT thesis series on BCSIR instruments: XRD, SEM, frequency- and temperature-dependent dielectric test

SELECTED PROJECTS

Millsorpt - Capstone Design Project · Department of MME, BUET

2025

- Converted ferrous mill scale, a low-value steel-industry waste, into a chemically treated adsorbent(hematite rich iron-oxide) for heavy metal wastewater treatment

- Designed a low-cost bench scale filtration column achieving a high removal of Cu, Cr and Ni from mixed-metal solutions; iron-oxide phases confirmed by XRD, metal removal quantified by AAS and UV-Vis.
- Presented the work to specialists at Abul Khair Steel, BCSIR, the Bangladesh Atomic Energy Commission etc., receiving positive technical feedback

3D PotterBot - CAD Modelling · SolidWorks

2023

- Modelled a paste-extrusion ceramic 3D printer as a fully-mated 118 parts assembly; produced photorealistic renders, an exploded-view motion study, and a static Finite Element Analysis of the vertical support rod

INDUSTRIAL AND LABORATORY TRAINING

Bangladesh Council of Scientific and Industrial Research (BCSIR) · Dhaka

2026

- Performed the characterization for my undergraduate thesis on BCSIR instruments: XRD, SEM, and temperature- and frequency-dependent dielectric measurement

Abul Khair Steel Melting Limited (AKSML) · Chittagong

March 2025

- Studied electric arc furnace, ladle refining furnace and continuous casting operations, and rolling-mill practice using Quenching & Tempering Bar (QTB) technology for structural steel.
- Observed quality-control workflows spanning ICP-OES, XRF, handheld XRF and wet chemical analysis, together with process automation and real-time monitoring

Bangladesh Industrial Technical Assistance Centre (BITAC) · Dhaka

March 2025

- Hands-on training in precision machining (lathe, milling, CNC), TIG/arc/spot welding, heat treatment, foundry operations, CAD-CAM and industrial automation

TECHNICAL SKILLS

- **Synthesis, Processing and Data Analysis:** Solid-state ceramic synthesis; hydrothermal synthesis; sintering; uniaxial pressing; XRD; FE-SEM and SEM; TEM including HRTEM, SAED; Impedance spectroscopy; frequency- and temperature-dependent dielectric measurement; d_{33} meter; cyclic voltammetry; I-V and Mott-Schottky analysis; UV-Vis spectroscopy; photoluminescence (PL); FTIR
- **Mechanical & Chemical Testing:** Tensile, Charpy impact and three-point flexural testing; Vickers, Rockwell and Brinell hardness; optical emission spectroscopy (OES); gravimetric and wet chemical analysis
- **Software:** OriginPro; MATLAB; SolidWorks; ImageJ; X'Pert HighScore; C/C++; MS Office; Adobe Photoshop and Illustrator

AWARDS AND SCHOLARSHIPS

- **University Merit Scholarship**, Bangladesh University of Engineering and Technology **(July 2022)**
- **University Stipend Scholarship**, Bangladesh University of Engineering and Technology (July 2021 and January 2022)
- **Letter of Appreciation**, Abul Khair Steel Melting Limited - for professionalism and work ethic during industrial training **2025**
- **Certificate of Perfect Attendance**, Notre Dame College, Dhaka **(2019 to 2020)**

LANGUAGE PROFICIENCY

English: IELTS Academic - Overall Band 7.0 (Listening 7.5, Reading 7.5, Writing 6.0, Speaking 6.5)

2026

LEADERSHIP AND EXTRACURRICULAR ACTIVITIES

- **Director of Administration**, Executive Committee, BUET Debating Club (BUETDC), 2025-26 :- previously Information Secretary. Organizing panel member for BUETDC Nationals 2024; finalist at the BUET Freshers' Debating Tournament; competed at the Inter-Faculty Tournament 2024 and the 6th GSCDC Nationals 2024.
- **Organizer**, Winter Fitness Competition 2025, BUET Gymnasium :- planned and ran the event in collaboration with the resident trainer; Organized Treasure-Hunt competition 2024, as an organizing panel member of "Reception'21"
- **Class Representative**, Department of MME, BUET, 2021-22 :- elected by classmates for both semesters of the first year
- **Active Member & Blood Donor**, Badhon - a non-profit voluntary blood donation organization, 2022-present